Hofmann–Streicher lifting of fibred categories

Dedicated to the memory of Thomas Streicher (1958–2025)

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***Abstract*—In 1997, Hofmann and Streicher introduced an explicit construction to lift a Grothendieck universe $\mathcal{U}$ from $\mathbf{Set}$ into the category of $\mathbf{Set}$-valued presheaves on a $\mathcal{U}$-small category B. More recently, Awodey presented an elegant functorial analysis of this construction in terms of the *categorical nerve*, the right adjoint to the functor that takes a presheaf to its category of elements; in particular, the categorical nerve's functorial action on the universal $\mathcal{U}$-small discrete fibration gives the generic family of $\mathcal{U}$'s Hofmann–Streicher lifting. Inspired by Awodey's analysis, we define a relative version of Hofmann–Streicher lifting in terms of the right pseudo-adjoint to the 2-functor $\mathbf{Fib}_A \to \mathbf{Fib}_B$ given by postcomposition with a fibration $p\colon A \twoheadrightarrow B$.**

***Index Terms*—universes, fibrations, Hofmann–Streicher lifting, 2-categories, pseudo-adjunctions.**

I. Introduction

Universes were originally introduced to axiomatic set theory by Zermelo [1] in 1930, and reinvented in the early 1960s by the Grothendieck School as a technical device for controlling the size of categories [2]. In the subsequent decade, several important developments occurred in short order:

- In 1971, Per Martin-Löf introduced his own version of universes in the context of his intuitionistic type theory [3], which Girard noticed was overly strong [4]. This led Martin-Löf to codify his now-standard presentation of (predicative) universes in type theory [5]–[7].
- Around the same time, Bénabou developed a categorical axiomatisation of universes [8] that makes sense for any elementary topos, subsuming both Zermelo and Grothendieck's set theoretic universes *and* Martin-Löf's type theoretic universes.

The type theoretic and category theoretic viewpoints on universes continued to develop in parallel. By the mid 1990s, Martin-Löf type theory had acquired several credible notions of *categorical model* [9]–[12]; this made it possible to consider natural questions, such as whether models of Martin-Löf type theory are stable under standard categorical constructions like presheaves. This question was answered in the affirmative by the late Martin Hofmann and Thomas Streicher in 1997, and the critical ingredient was what we now refer to as the *Hofmann–Streicher universe lifting* construction [13]:

> Given a universe $p\colon \tilde{\mathcal{U}} \to \mathcal{U}$ in $\mathbf{Set}$ and a $\mathcal{U}$-small category C, there is a universe $p_{\hat{C}}\colon \tilde{\mathcal{U}}_{\hat{C}} \to \mathcal{U}_{\hat{C}}$ in $\hat{C}$ defined as follows:
>
> $$\begin{aligned}\tilde{\mathcal{U}}_{\hat{C}}(c) &= \mathbf{Cat}((C/c)^{op}, \tilde{\mathcal{U}})\\ \mathcal{U}_{\hat{C}}(c) &= \mathbf{Cat}((C/c)^{op}, \mathcal{U})\\ p^c_{\hat{C}}(x) &= (f\colon d \to c) \mapsto p(x(f))\end{aligned}$$

Hofmann and Streicher's lifting construction does not depend on the axioms of universes, but it does preserve them. In fact, the construction itself requires only a full internal subcategory of $\mathbf{Set}$ presented by a generic family.

A. Hofmann—Streicher lifting and universal properties

Although universes in the sense of Bénabou [8], Streicher [14], and Martin-Löf [5] are not defined by means of a universal property, Hofmann–Streicher lifting does preserve the universal properties of more restrictive kinds of universes, including Lawvere's subobject classifiers. Indeed, when applied to the full internal subcategory of *propositions* in $\mathbf{Set}$ which is presented by the subobject classifier $\{1\} \hookrightarrow \mathbf{2}$, the Hofmann–Streicher lifting construction yields the subobject classifier of any presheaf topos; indeed, we have:

$$\Omega_{\hat{C}}(c) = \mathbf{Cat}((C/c)^{op}, \mathbf{2}) = \mathrm{Sieve}_C(c)$$

Remark 1: There is a subtlety here, namely that a category C is never $\mathbf{2}$-small unless it is either the initial or terminal category; nonetheless, the Hofmann–Streicher lifting does exist and is the subobject classifier of $\hat{C}$. In fact, the assumption that C is $\mathcal{U}$-small was needed by Hofmann and Streicher only to close the lifted universe under type theoretic connectives such as function spaces; closure of $\Omega_{\hat{C}}$ under logical connectives follows by different means.

The universal property of the subobject classifier is 1-dimensional and can therefore be phrased in the language of 1-toposes. The corresponding notion in a 2-dimensional topos would then classify (internal) *discrete fibrations* rather than subobjects, as described by Weber [15]. In the 2-topos $\mathbf{Cat}$ of categories, the classifying discrete opfibration is of course $\partial_1\colon \mathbf{1}/\mathbf{Set} \to \mathbf{Set}$.

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Although Weber [15] seems to have been unaware of Hofmann and Streicher's universe lifting construction, he appears to invent it independently. He applies it to the discrete opfibration classifier $\mathbf{1}/\mathbf{Set} \to \mathbf{Set}$ of the 2-topos $\mathbf{Set}$ to obtain the discrete opfibration classifier of the (strict) functor 2-topos $\mathbf{Cat}(\hat{C})$ consisting of strict functors $C^{op} \to \mathbf{Cat}$, natural transformations between them, and modifications between those. Of course, $\mathbf{Cat}(\hat{C})$ is 2-equivalent to the 2-category of *split* fibrations, with functors strictly preserving the splittings, and natural transformations between these.

B. Functoriality of Hofmann–Streicher lifting

Awodey [16] recently showed that Hofmann and Streicher's original universe lifting construction is part of a 1-functor $\nu_C \colon \mathbf{cat}_s \to \hat{C}$ that he calls the *categorical nerve*, where we have written $\mathbf{cat}_s$ for the 1-category of small categories and functors taken up to equality.

Two of Awodey's crucial insights are as follows:

1) The categorical nerve $\nu_C \colon \mathbf{cat}_s \to \hat{C}$ emerges as the right adjoint to the category of elements functor $\int_C \colon \hat{C} \to \mathbf{cat}_s$. In particular, Awodey defines:
$$\nu_C(D)(c) = \mathbf{cat}_s(C/c, D)$$
2) The Hofmann–Streicher lifting of a universe $\tilde{\mathcal{U}} \to \mathcal{U}$ in $\mathbf{Set}$ to $\hat{C}$ is obtained by applying the categorical nerve to the arrow $\tilde{\mathcal{U}}^{op} \to \mathcal{U}^{op}$ in $\mathbf{cat}_s$ which is the classifying discrete *fibration* in $\mathbf{cat}$:
$$\begin{aligned}&\nu_C(\mathcal{U}^{op})(c)\\ &\quad = \mathbf{cat}_s(C/c, \mathcal{U}^{op})\\ &\quad = \mathbf{cat}_s((C/c)^{op}, \mathcal{U})\\ &\quad = \mathcal{U}_{\hat{C}}(c)\end{aligned}$$

In hindsight, it can be seen that something very similar is happening in Weber's construction [15] of the classifying discrete opfibration of $\mathbf{Cat}(\hat{C})$. In particular, Weber considers the following 2-adjunction in which $E_C(X) = \left(\int_C X^{op}\right)^{op}$ and $\mathrm{Sp}_C(D)(c) = \mathbf{Cat}((C/c)^{op}, D)$:
$$\begin{tikzcd} \mathbf{Cat} \arrow[r, shift right, "\mathrm{Sp}_C"'] \arrow[r, phantom, "\perp"] & \mathbf{Cat}(\hat{C}) \arrow[l, shift right, "E_C"'] \end{tikzcd}$$

To properly compare Awodey's and Weber's constructions, we first factor Weber's construction through the duality involutions to reveal a 2-dimensional version of Awodey's categorical nerve, which we write $\int_C \dashv \nu_C \colon \mathbf{Cat} \to \mathbf{Cat}(\hat{C})$:
$$\begin{tikzcd} \mathbf{Cat} \arrow[r, shift right, "op"'] \arrow[r, phantom, "\perp"] & \mathbf{Cat}^{co} \arrow[l, shift right, "op"'] \arrow[r, shift right, "\nu_C^{co}"'] \arrow[r, phantom, "\perp"] & \mathbf{Cat}(\hat{C})^{co} \arrow[l, shift right, "\int_C^{co}"'] \arrow[r, shift right, "op"'] \arrow[r, phantom, "\perp"] & \mathbf{Cat}(\hat{C}) \arrow[l, shift right, "op"'] \end{tikzcd}$$

The Grothendieck construction $\int_C \colon \mathbf{Cat}(\hat{C}) \to \mathbf{Cat}$ takes a presheaf of categories to its oplax colimit; its conjugation by duality involutions $op \circ \int_C^{co} \circ op$ therefore takes a presheaf of categories to its *lax* colimit. Therefore, the 2-dimensional version $\nu_C \colon \mathbf{Cat} \to \mathbf{Cat}(\hat{C})$ is some kind of "oplax diagonal 2-functor" and therefore the Hofmann–Streicher lifting 2-functor is some kind of "lax diagonal 2-functor".

C. Our contributions

In this paper, we expose a further generalisation of Hofmann–Streicher lifting. In particular, we describe a *relative* version of Hofmann–Streicher lifting defined in terms of a right *pseudo*-adjoint to the postcomposition functor $(p \circ -) \colon \mathbf{Fib}_A \to \mathbf{Fib}_B$ for a given fibration $p \colon A \to B$, where $\mathbf{Fib}_C$ is the 2-category of fibred categories over C, fibred functors over C, and fibred natural transformations between them.

We therefore differ from the prior work in main two ways: after replacing $\mathbf{Cat}(\hat{C})$ with $\mathbf{Fib}_C$, we are then able to relativise the total category 2-functor $\int_A \colon \mathbf{Fib}_A \to \mathbf{Cat}$ to the more general postcomposition 2-functor $(p \circ -) \colon \mathbf{Fib}_A \to \mathbf{Fib}_B$ induced by any fibration $p \colon A \to B$. Indeed, under the identification $\mathbf{Cat} \cong \mathbf{Fib}_{\mathbf{1}}$, the total category 2-functor is precisely the postcomposition with the universal functor $A \to \mathbf{1}$. The fibrational relativisation of Hofmann–Streicher lifting makes it possible for the first time to consider *iterated* lifting of universes and other structures.

1) From split fibrations to fibrations: The most direct generalisation of Weber's construction would have yielded a 2-functor $\mathbf{Cat}(\hat{B}) \to \mathbf{Cat}(\hat{A})$ but it is ultimately more useful to work with fibrations. Indeed, $\mathbf{Cat}(\hat{C})$ is 2-equivalent to the 2-category $\mathbf{sFib}_C$ of split fibrations over C, split fibred functors over C, and fibred natural transformations between them. Unfortunately, the evident forgetful 2-functor $U \colon \mathbf{sFib}_C \to \mathbf{Fib}_C$ is not even a bi-equivalence, although Giraud exhibits both left and right 2-adjoints [17]. Therefore, although it does not matter in the end whether one asks for fibrations or split fibrations, it is overly restrictive in practice to ask that these splittings be preserved on the nose by morphisms. This is a special case of a familiar pattern in higher categorical algebra, where one prefers to consider *weak* homomorphisms regardless of whether the algebras are strict or weak.

2) Relativising along a fibration $p \colon A \to B$: As it stands, the Hofmann–Streicher construction cannot be iterated. For example, we may take a universe $\mathcal{U} \in \mathbf{Set}$ of sets to a universe $\mathcal{U}_{\hat{C}} \in \hat{C}$ of presheaves; but if D is an internal category in $\hat{C}$, we do not have any way to lift $\mathcal{U}_{\hat{C}}$ to a universe of internal presheaves on D over $\hat{C}$. To solve this problem, we *relativise* the Hofmann–Streicher construction so that it goes not from $\mathbf{Cat}$ to $\mathbf{Fib}_A$ but rather from $\mathbf{Fib}_B$ to $\mathbf{Fib}_A$, parameterised in a fibration $p \colon A \to B$. When $B = \mathbf{1}$, we recover the usual Hofmann–Streicher lifting.

To see how relativisation helps with iteration, we return to our example of a universe $\mathcal{U}$ of sets, which we can view as a category of $\mathcal{U}$-small sets and functions between them. Given a small category C, we can lift $\mathcal{U}$ to a fibred category $\mathcal{U}_C$ over C using the ordinary Hofmann–Streicher lifting. Next, an internal category $D \in \hat{C}$ can be viewed instead as a fibration $p_D \colon D \to C$; our *relative* Hofmann–Streicher lifting of $\mathcal{U}_C \in \mathbf{Fib}_C$ along the fibration $p_D \colon D \to C$ produces, then, a fibration $\mathcal{U}_D \in \mathbf{Fib}_D$ which is the desired universe of internal presheaves on D over $\hat{C}$. Here we are using the standard observation that when $p \colon D \to C$ is a fibration, a further fibration $q \colon E \to D$ can equally well be viewed as a fibration in $\mathbf{Cat}$ over D *or*

as an internal fibration in $\mathbf{Fib}_C$ over p.

If ordinary Hofmann–Streicher lifting is the needful tool for defining presheaf models of Martin-Löf type theory, then the relative Hofmann–Streicher lifting is precisely what is needed to define *internal* presheaf models of Martin-Löf type theory within another presheaf model. In fact, this arises more often than one might at first guess: for example, Bizjak *et al.* [18] describe what *ought* to amount to the standard presheaf model of cubical type theory [19], [20] internal to the presheaf model of guarded dependent type theory [21], [22], but lacking the tools developed in this paper it was necessary to simultaneously describe the guarded and cubical aspects.

3) Relative Hofmann–Streicher lifting as a pseudo-adjoint: Given a fibration $p\colon A \twoheadrightarrow B$, our *relative* Hofmann–Streicher lifting takes the form of a 2-functor $\mathbf{Fib}_B \to \mathbf{Fib}_A$, which is right *pseudo*-adjoint to the 2-functor $(p \circ -)\colon \mathbf{Fib}_A \to \mathbf{Fib}_B$ that sends a fibration $q\colon E \to A$ to the composite fibration $p \circ q\colon E \to B$.

II. Technical overview

In this section, we provide a high-level overview of the mathematical content of the paper; this can be read by someone who wishes to understand the overall architecture of the generalised Hofmann–Streicher lifting and its applications, without necessarily grappling with all the details. For the full technical development, see § III.

A. Displayed and fibred categories

We presuppose some knowledge of fibred category theory. On top of this, we will at times employ a convention that assists with remembering when a particular fibration is to be thought of a special kind of parameterised category *vs.* as a mere functor satisfying some properties.

1) We will refer to objects of $\mathbf{Cat}/B$ and $\mathbf{Fib}_B$ as "displayed" and "fibred" categories to emphasise their role as objects of a 2-category of category-like things.
2) Given a displayed category $E \in \mathbf{Cat}/B$, we will write $B.E$ for its total category and we shall refer to the projection functor $p_{B.E}\colon B.E \to B$ as its *display*.
3) Given a fibred category $E \in \mathbf{Fib}_B$, we shall refer to the projection $p_{B.E}\colon B.E \twoheadrightarrow B$ as the corresponding *fibration*.

The specifics of how to represent displayed and fibred categories do not play a role in this paper; some readers may prefer to think of them as represented by their underlying display functors, whereas other readers may prefer to think of them in the terms advocated by Ahrens and Lumsdaine [23]. We take the latter viewpoint in our technical development, but it is not mandatory.

B. Base change and sum of displayed categories

Let $p\colon A \to B$ be an arbitrary functor. There is a *change of base* 2-functor $\Delta_p\colon \mathbf{Cat}/B \to \mathbf{Cat}/A$ computed on display functors by *pseudo-pullback* along p; the change of base 2-functor has a left pseudo-adjoint $\Sigma_p\colon \mathbf{Cat}/A \to \mathbf{Cat}/B$ computed on display functors by *postcomposition* with p. As left pseudo-adjoint to change of base, Σ_p deserves to be called the *sum* along p.

C. Base change and sum of fibred categories

For any functor $p\colon A \to B$, the base change 2-functor $\Delta_p\colon \mathbf{Cat}/B \to \mathbf{Cat}/A$ restricts to a base change 2-functor $p^*\colon \mathbf{Fib}_B \to \mathbf{Fib}_A$ as follows:[1]

$$\begin{tikzcd} \mathbf{Fib}_B \arrow[r, dashed, "p^*"] \arrow[d, "U_B"'] & \mathbf{Fib}_A \arrow[d, "U_A"] \\ \mathbf{Cat}/B \arrow[r, "\Delta_p"'] & \mathbf{Cat}/A \end{tikzcd}$$

The base change 2-functor $p^*\colon \mathbf{Fib}_B \to \mathbf{Fib}_A$ always has a left pseudo-adjoint $p_!\colon \mathbf{Fib}_A \to \mathbf{Fib}_B$ that deserves to be called the *sum (of fibrations) along* p, but this does *not* factor through the sum of Σ_p of displayed categories. Indeed, this can be seen immediately because if $p\colon A \to B$ is not a fibration, then the sum of the underlying displayed category of a fibred category E along p need not be a fibred category.

Interpreting Bénabou, Streicher [24] explains that when $p\colon A \to \mathbf{1}$ is the punctual map, the sum of fibrations $p_!E$ is the localisation of $\Sigma_p(E)$ that inverts all cartesian arrows; put more simply, $p_!E$ is the *pseudo-colimit* of the pseudofunctor corresponding to E under straightening (which we will denote by $E_\bullet\colon A^{op} \to \mathbf{Cat}$), whereas $\Sigma_p(E)$ is the *oplax colimit* of the same pseudofunctor.

D. Oplax sum and base change of fibred categories

Whenever $p\colon A \twoheadrightarrow B$ is a fibration, the sum of displayed categories along p restricts to a 2-functor on fibred categories that computes what we would most rightly call the "oplax sum" along p.

$$\begin{tikzcd} \mathbf{Fib}_A \arrow[r, dashed, "\Sigma_p^{opl}"] \arrow[d, "U_A"'] & \mathbf{Fib}_B \arrow[d, "U_B"] \\ \mathbf{Cat}/A \arrow[r, "\Sigma_p"'] & \mathbf{Cat}/B \end{tikzcd}$$

Streicher [24, Theorem 4.1] points out that in the case of a fibred category $E \in \mathbf{Fib}_A$, the displayed projection functor $\Sigma_p^{opl}(E) \to A$ over B is fibred in the sense of preserving cartesian arrows. This tells us that cartesian lifts in $\Sigma_p^{opl}(E)$ are computed successively using cartesian lifts in A over B and in E over A.

Our main technical result is to show that the oplax sum $\Sigma_p^{opl}\colon \mathbf{Fib}_A \to \mathbf{Fib}_B$ has a *right* pseudo-adjoint $\Delta_p^{opl}\colon \mathbf{Fib}_B \to \mathbf{Fib}_A$ which can be used to compute a generalisation of Hofmann and Streicher's universe construction. We are forced by convention, it would seem, to refer to Δ_p^{opl} as the *oplax base change* of fibred categories along p.

The fibred Yoneda lemma gives a quick heuristic to compute $\Delta_p^{opl}(F)$ as a split fibration for any $F \in \mathbf{Fib}_B$ as follows (where we have written $\mathbf{y}_A\colon A \hookrightarrow \mathbf{Fib}_A$ for the functor that

[1] In fact, p^* can be computed on fibrations in terms of strict pullbacks if you like, because a strict pullback of a fibration is always a pseudo-pullback.

sends $a \in A$ to the discrete domain fibration $\partial_1 \colon A/a \to A$): its straightening $\Delta_p^{opl}(F)_\bullet \colon A^{op} \to \mathbf{Cat}$ should satisfy

$$\Delta_p^{opl}(F)_\bullet \cong \mathbf{Fib}_B(\Sigma_p^{opl}(\mathbf{y}_A(-)), F)$$

Of course, one still must check that this construction really does yield a right pseudo-adjoint to Σ_p^{opl}.

E. Lax sum and base change of fibred categories

By conjugating with duality involutions, we can obtain *lax* versions of sum and base change for fibred categories. In particular, if we have the pseudo-adjunction $\Sigma_p^{opl} \dashv \Delta_p^{opl} \colon \mathbf{Fib}_B \to \mathbf{Fib}_A$, we obtain a corresponding pseudo-adjunction $\Sigma_p^{lax} \dashv \Delta_p^{lax} \colon \mathbf{Fib}_B \to \mathbf{Fib}_A$ as follows:

$$\begin{tikzcd}
& \mathbf{Fib}_A^{co} \arrow[r, "(\Sigma_p^{opl})^{co}"] & \mathbf{Fib}_B^{co} \arrow[dr, "op"] & \\
\mathbf{Fib}_A \arrow[ur, "op"] \arrow[rrr, dashed, bend left, "\Sigma_p^{lax}"] \arrow[rrr, dashed, bend right, "\Delta_p^{lax}"'] \arrow[rrr, phantom, "\perp"] & & & \mathbf{Fib}_B \arrow[dl, "op"] \\
& \mathbf{Fib}_A^{co} \arrow[ul, "op"] & \mathbf{Fib}_B^{co} \arrow[l, "(\Delta_p^{lax})^{co}"] &
\end{tikzcd}$$

F. Hofmann–Streicher lifting as lax base change

Let $p \colon A \twoheadrightarrow B$ be a fibration, and let $E \in \mathbf{Fib}_B$ be a fibred category over B; then we can finally define the generalised Hofmann–Streicher lifting of E along p to be the *lax base change* $\Delta_p^{lax}(E) \in \mathbf{Fib}_A$. In order to explain why this deserves the name, we consider the case where $!_A \colon A \twoheadrightarrow \mathbf{1}$ is the punctual map; for any category $E \in \mathbf{Cat} \cong \mathbf{Fib_1}$, we compute the fibres of the lax base change $\Delta_{!_A}^{lax}(E) \in \mathbf{Fib}_A$:

$$\begin{aligned}
&\Delta_{!_A}^{lax}(E)_\bullet \\
&\quad = (\Delta_{!_A}^{opl}(E^{op}))_\bullet^{op} \\
&\quad = (\Delta_{!_A}^{opl}(E^{op})_\bullet)^{op} \\
&\quad \cong \mathbf{Cat}(\Sigma_{!_A}^{opl}(\mathbf{y}_A(-)), E^{op})^{op} \\
&\quad = \mathbf{Cat}(A/-, E^{op})^{op} \\
&\quad = \mathbf{Cat}((A/-)^{op}, E)
\end{aligned}$$

Now let $\mathcal{U}$ be a Grothendieck universe presented by a generic family of $\mathcal{U}$-small sets $\pi_{\mathcal{U}} \colon \tilde{\mathcal{U}} \to \mathcal{U}$; we now assume that A is $\mathcal{U}$-small. The generic family of $\mathcal{U}$ is the image in $\mathbf{Set}$ under $\mathrm{ob} \colon \mathbf{Cat} \to \mathbf{Set}$ of the forgetful functor $\partial_1 \colon \mathbf{1}/\mathbf{Set}_{\mathcal{U}} \to \mathbf{Set}_{\mathcal{U}}$ from pointed $\mathcal{U}$-small sets to $\mathcal{U}$-small sets. Applying the functorial action of $\Delta_{!_A}^{lax} \colon \mathbf{Cat} \to \mathbf{Fib}_A$ to this forgetful functor, followed by that of $\mathrm{ob} \colon \mathbf{Fib}_A \to \mathbf{DFib}_A \cong \hat{A}$ gives us *precisely* the generic family of Hofmann and Streicher's lifted presheaf universe:

$$\begin{aligned}
&\mathrm{ob}(\mathbf{Cat}((A/-)^{op}, \partial_1)) \\
&\quad : \mathrm{ob}(\mathbf{Cat}((A/-)^{op}, \mathbf{1}/\mathbf{Set}_{\mathcal{U}})) \\
&\quad \to \mathrm{ob}(\mathbf{Cat}((A/-)^{op}, \mathbf{Set}_{\mathcal{U}}))
\end{aligned}$$

III. Technical development

We assume that all fibred categories are cloven, but not necessarily split; naturally, we do not require cleavings to be preserved by fibred functors. The impact of this choice depends on the foundational system in which this paper is interpreted:

1) In foundations where the axiom of choice holds, any fibred category can be equipped with a cleaving.[2] Thus the forgetful 2-functor from cloven fibred categories to uncloven fibred categories would be a biequivalence.
2) In foundations where choice may not hold (*e.g.* a topos), the correct notion of fibred category is usually the cloven one anyway, which agrees with Chevalley's definition.
3) In univalent foundations, a (univalent) fibred category is *uniquely* cloven [23], so the question doesn't arise.

We will use the following especially verbose notation for the components of the given cleaving of a fibred category E:

$$\begin{tikzcd}
\ell_E(b_{01}, e) \arrow[r, "\bar{\ell}_E(b_{01}, e)"] \arrow[d, mapsto] \arrow[dr, phantom, "\lrcorner"] & e \arrow[d, mapsto] & & B.E \arrow[d] \\
b_0 \arrow[r, "b_{01}"'] & b_1 & & B
\end{tikzcd}$$

A. 2-functoriality of oplax base change

In this section, we will define an "oplax base change" 2-functor $\Delta_p^{opl} \colon \mathbf{Fib}_B \to \mathbf{Fib}_A$ for any fibration $p \colon A \twoheadrightarrow B$.

1) Action of oplax base change on on 0-cells: Let $E \in \mathbf{Fib}_B$ be a fibred category over B. We first define a displayed category $\Delta_p^{opl}(E)$ over A called the *oplax base change of E along p* as follows.

- A displayed object of $\Delta_p^{opl}(E)$ over $a \in A$ is defined to be a fibred functor $e \colon \Sigma_p^{opl}(\mathbf{y}_A(a)) \to E$ over B.
- A displayed arrow from e_0 to e_1 over $a_{01} \colon a_0 \to a_1$ is given by a 2-cell $e_{01} \colon e_0 \to e_1 \circ \Sigma_p^{opl}(\mathbf{y}_A(a_{01}))$ in $\mathbf{Fib}_B(\Sigma_p^{opl}(\mathbf{y}_A(a_0)), E)$ as depicted below.

$$\begin{tikzcd}
\Sigma_p^{opl}(\mathbf{y}_A(a_0)) \arrow[r, "e_0"] \arrow[d, "\Sigma_p^{opl}(\mathbf{y}_A(a_{01}))"'] \arrow[dr, phantom, "\Rightarrow e_{01}"] & E \\
\Sigma_p^{opl}(\mathbf{y}_A(a_1)) \arrow[ur, "e_1"'] &
\end{tikzcd}$$

- The displayed identity arrow $1_e \colon e \to e$ over $a \in A$ is the corresponding identity arrow $1_e \colon e \to e \equiv e \circ \Sigma_p^{opl}(\mathbf{y}_A(1_a))$.
- Given displayed arrows $e_{01} \colon e_0 \to e_1$ and $e_{12} \colon e_1 \to e_2$ over $a_{01} \colon a_0 \to a_1$ and $a_{12} \colon a_1 \to a_2$, the displayed composite $e_{12} \circ e_{01} \colon e_0 \to e_2$ over $a_{12} \circ a_{01}$ is given by the following composite 2-cell:

$$\begin{tikzcd}
\Sigma_p^{opl}(\mathbf{y}_A(a_0)) \arrow[r, "e_0"] \arrow[d, "\Sigma_p^{opl}(\mathbf{y}_A(a_{01}))"'] \arrow[dr, phantom, "\Rightarrow e_{01}"] & E \\
\Sigma_p^{opl}(\mathbf{y}_A(a_1)) \arrow[ur, "e_1"'] \arrow[d, "\Sigma_p^{opl}(\mathbf{y}_A(a_{12}))"'] \arrow[dr, phantom, "\Rightarrow e_{12}"] & \\
\Sigma_p^{opl}(\mathbf{y}_A(a_2)) \arrow[uur, bend right, "e_2"'] &
\end{tikzcd}$$

[2] Technically, one would need the axiom of choice for *classes* or *large sets*.

Lemma 2: Let $E \in \mathbf{Fib}_B$ be a fibred category over B, and let $p\colon A \twoheadrightarrow B$ be a fibration. In $\Delta_p^{opl}(E)$, a displayed arrow $e_{01}\colon e_0 \to e_1$ over $a_{01}\colon a_0 \to a_1$ is *cartesian* if and only if the underlying 2-cell $e_{01}\colon e_0 \to e_1 \circ \Sigma_p^{opl}(\mathbf{y}_A(a_{01}))$ is invertible in $\mathbf{Fib}_B(\Sigma_p^{opl}(\mathbf{y}_A(a_0)), E)$.

Corollary 3: Let $E \in \mathbf{Fib}_B$ be a fibred category over B with $p\colon A \twoheadrightarrow B$ a fibration. Then the displayed category $\Delta_p^{opl}(E)$ can be *split fibred* over A by defining

$$\begin{tikzcd}
\ell_{\Delta_p^{opl}(E)}(a_{01}, e) \arrow[r, "\bar{\ell}_{\Delta_p^{opl}(E)}(a_{01},e)"] \arrow[d, mapsto] & e \arrow[d, mapsto] \\
a_0 \arrow[r, "a_{01}"'] & a_1
\end{tikzcd}$$

to be the following identity 2-cell, setting $\ell_{\Delta_p^{opl}(E)}(a_{01}, e) = e \circ \Sigma_p^{opl}(\mathbf{y}_A(a_{01}))$:

$$\begin{tikzcd}
\Sigma_p^{opl}(\mathbf{y}_A(a_0)) \arrow[r, "\ell_{\Delta_p^{opl}(E)}(a_{01},e)"] \arrow[d, "\Sigma_p^{opl}(\mathbf{y}_A(a_{01}))"'] & E \\
\Sigma_p^{opl}(\mathbf{y}_A(a_1)) \arrow[ur, "e"'] &
\end{tikzcd}$$

The 2-functor $\Delta_p^{opl}(E)_\bullet\colon A^{op} \to \mathbf{Cat}$ corresponding to the split cleaving specified in Corollary 3 is precisely the one foreshadowed in § II-D:

$$\Delta_p^{opl}(E)_\bullet = \mathbf{Fib}_B(\Sigma_p^{opl}(\mathbf{y}_A(-)), E)$$

2) Action of oplax base change on 1- and 2-cells: Let E and F be two fibred categories over B. In this section, we construct the local hom functors $(\Delta_p^{opl})_{E,F}\colon \mathbf{Fib}_B(E,F) \to \mathbf{Fib}_A(\Delta_p^{opl}(E), \Delta_p^{opl}(F))$.

a) Action of local hom functor on 0-cells: Let $f\colon E \to F$ be a fibred functor over B; we define a displayed functor $(\Delta_p^{opl})_{E,F}(f)\colon \Delta_p^{opl}(E) \to \Delta_p^{opl}(F)$ as follows:

- A displayed object $e \in \Delta_p^{opl}(E)$ over $a \in A$ is sent to the composite fibred functor $f \circ e\colon \Sigma_p^{opl}(\mathbf{y}_A(a)) \to F$.
- A displayed arrow $e_{01}\colon e_0 \to e_1$ in $\Delta_p^{opl}(E)$ over $a_{01}\colon a_0 \to a_1$ is sent to its whiskering with f as shown:

$$\begin{tikzcd}
\Sigma_p^{opl}(\mathbf{y}_A(a_0)) \arrow[r, "e_0"] \arrow[d, "\Sigma_p^{opl}(\mathbf{y}_A(a_{01}))"'] \arrow[dr, phantom, "\Rightarrow e_{01}"] & E \arrow[r, "f"] & F \\
\Sigma_p^{opl}(\mathbf{y}_A(a_1)) \arrow[ur, "e_1"'] & &
\end{tikzcd}$$

Lemma 4: The displayed functor $(\Delta_p^{opl})_{E,F}(f)\colon \Delta_p^{opl}(E) \to \Delta_p^{opl}(F)$ is split fibred in the sense that it preserves the cartesian lifts specified in Corollary 3 on the nose.

b) Action of local hom functor on 1-cells: Let $f_{01}\colon f_0 \to f_1$ be a vertical natural transformation of fibred functors $f_0, f_1\colon E \to F$ over B. The component of the functorial action $(\Delta_p^{opl})_{E,F}^{f_0,f_1}(f_{01})\colon (\Delta_p^{opl})_{E,F}(f_0) \to (\Delta_p^{opl})_{E,F}(f_1)$ at $e \in \Delta_p^{opl}(E)$ over $a \in A$ is the following composite 2-cell:

$$\begin{tikzcd}
\Sigma_p^{opl}(\mathbf{y}_A(a)) \arrow[r, "e"] \arrow[d, equal] & E \arrow[r, bend left, "f_0"] \arrow[r, bend right, "f_1"'] \arrow[r, phantom, "\Downarrow f_{01}"] & F \\
\Sigma_p^{opl}(\mathbf{y}_A(a)) \arrow[ur, "e"'] & &
\end{tikzcd}$$

We further verify the following naturality square for each $e_{01}\colon e_0 \to e_1$ in $\Delta_p^{opl}(E)$ over $a_{01}\colon a_0 \to a_1$:

$$\begin{tikzcd}
(\Delta_p^{opl})_{E,F}(f_0)(e_0) \arrow[r, "(\Delta_p^{opl})_{E,F}^{f_0,f_1}(f_{01})_{e_0}"] \arrow[d, "(\Delta_p^{opl})_{E,F}(f_0)(e_{01})"'] & (\Delta_p^{opl})_{E,F}(f_1)(e_0) \arrow[d, "(\Delta_p^{opl})_{E,F}(f_1)(e_{01})"'] \\
(\Delta_p^{opl})_{E,F}(f_0)(e_1) \arrow[r, "(\Delta_p^{opl})_{E,F}^{f_0,f_1}(f_{01})_{e_1}"'] & (\Delta_p^{opl})_{E,F}(f_1)(e_1)
\end{tikzcd}$$

That this square commutes follows from the interchange of the 2-cells f_{01} and e_{01} in the pasting diagrams representing the two composites.

We have now constructed each of the local hom functors $(\Delta_p^{opl})_{E,F}\colon \mathbf{Fib}_B(E,F) \to \mathbf{Fib}_A(\Delta_p^{opl}(E), \Delta_p^{opl}(F))$.

3) Respect of local hom functors for composition and identity: In order to conclude that the actions of oplax base change on 0-, 1-, and 2-cells that we have defined above assemble into a 2-functor, it remains to check that the assignment of local hom functors respects horizontal composition and identity data. In what follows, we shall write $\Gamma_{\mathfrak{C}}$ and $\mathrm{J}_{\mathfrak{C}}$ for the respective horizontal composition and identity data of a 2-category $\mathfrak{C}$.

Lemma 5: For any three fibred categories $E, F, G \in \mathbf{Fib}_B$ over B, the square depicted in Figure 1 commutes strictly.

Lemma 6: Let $p\colon A \twoheadrightarrow B$ be a fibration. For any fibred category $E \in \mathbf{Fib}_B$, the following triangle commutes strictly:

$$\begin{tikzcd}
\mathbf{1}_{\mathbf{Cat}} \arrow[r, "\mathrm{J}^E_{\mathbf{Fib}_B}"] \arrow[dr, "\mathrm{J}^{\Delta_p^{opl}(E)}_{\mathbf{Fib}_A}"'] & \mathbf{Fib}_B(E,E) \arrow[d, "(\Delta_p^{opl})_{E,E}"] \\
& \mathbf{Fib}_A(\Delta_p^{opl}(E), \Delta_p^{opl}(E))
\end{tikzcd}$$

Corollary 7: The assignments $E \mapsto \Delta_p^{opl}(E)$, $f \mapsto (\Delta_p^{opl})_{E,F}(f)$, and $f_{01} \mapsto (\Delta_p^{opl})_{E,F}^{f_0,f_1}(f_{01})$ assemble into a 2-functor $\Delta_p^{opl}\colon \mathbf{Fib}_B \to \mathbf{Fib}_A$ for any fibration $p\colon A \twoheadrightarrow B$.

B. Oplax base change as a right pseudo-adjoint

Let $p\colon A \twoheadrightarrow B$ be a fibration. We plan to exhibit the oplax base change 2-functor $\Delta_p^{opl}\colon \mathbf{Fib}_B \to \mathbf{Fib}_A$ as the right pseudo-adjoint to the oplax sum $\Sigma_p^{opl}\colon \mathbf{Fib}_A \to \mathbf{Fib}_B$. There are multiple ways to present pseudo-adjunctions, but we will prefer the one based on a pseudo-natural equivalence of hom-categories. In other words:

- We must assign to fibred categories $E \in \mathbf{Fib}_A$ and $F \in \mathbf{Fib}_B$ equivalences $\sharp_{E,F}\colon \mathbf{Fib}_A(E, \Delta_p^{opl}(F)) \simeq \mathbf{Fib}_B(\Sigma_p^{opl}(E), F)$.

$$\begin{tikzcd}
\mathbf{Fib}_B(E,F) \times \mathbf{Fib}_B(F,G) \arrow[r, "\Gamma^{E,F,G}_{\mathbf{Fib}_B}"] \arrow[d, "(\Delta_p^{opl})_{E,F} \times (\Delta_p^{opl})_{F,G}"'] & \mathbf{Fib}_B(E,G) \arrow[d, "(\Delta_p^{opl})_{E,G}"] \\
\mathbf{Fib}_A(\Delta_p^{opl}(E), \Delta_p^{opl}(F)) \times \mathbf{Fib}_A(\Delta_p^{opl}(F), \Delta_p^{opl}(G)) \arrow[r, "\Gamma^{\Delta_p^{opl}(E),\Delta_p^{opl}(F),\Delta_p^{opl}(G)}_{\mathbf{Fib}_A}"'] & \mathbf{Fib}_A(\Delta_p^{opl}(E), \Delta_p^{opl}(G))
\end{tikzcd}$$

Fig. 1. Preservation of horizontal composition by the local hom functors of oplax base change.

- To fibred functors $E_{10} \colon E_1 \to E_0$ and $F_{01} \colon F_0 \to F_1$ we must assign invertible pseudo-naturality cells as follows:

$$\begin{tikzcd}
\mathbf{Fib}_A(E_0, \Delta_p^{opl}(F_0)) \arrow[r, "\sharp_{E_0,F_0}"] \arrow[d, "\mathbf{Fib}_A(E_{10},\Delta_p^{opl}(F_{01}))"'] \arrow[dr, phantom, "\Downarrow \sharp_{E_{10},F_{01}}"] & \mathbf{Fib}_B(\Sigma_p^{opl}(E_0), F_0) \arrow[d, "\mathbf{Fib}_B(\Sigma_p^{opl}(E_{10}),F_{01})"] \\
\mathbf{Fib}_A(E_1, \Delta_p^{opl}(F_1)) \arrow[r, "\sharp_{E_1,F_1}"'] & \mathbf{Fib}_B(\Sigma_p^{opl}(E_1), F_1)
\end{tikzcd}$$

- The pseudo-naturality datum $\sharp_{1_E,1_F}$ needs to be equal to the identity 2-cell on $\sharp_{E,F} \colon \mathbf{Fib}_A(E, \Delta_p^{opl}(F)) \to \mathbf{Fib}_B(\Sigma_p^{opl}(E), F)$.
- For $E_{21} \colon E_2 \to E_1$, $E_{10} \colon E_1 \to E_0$, $F_{01} \colon E_0 \to E_1$ and $F_{12} \colon E_1 \to E_2$, the pseudonaturality datum $\sharp_{E_{10} \circ E_{21}, F_{12} \circ F_{01}}$ must be equal to the following pasting:

$$\begin{tikzcd}
\mathbf{Fib}_A(E_0, \Delta_p^{opl}(F_0)) \arrow[r, "\sharp_{E_0,F_0}"] \arrow[d, "\mathbf{Fib}_A(E_{10},\Delta_p^{opl}(F_{01}))"'] \arrow[dr, phantom, "\Downarrow \sharp_{E_{10},F_{01}}"] & \mathbf{Fib}_B(\Sigma_p^{opl}(E_0), F_0) \arrow[d, "\mathbf{Fib}_B(\Sigma_p^{opl}(E_{10}),F_{01})"] \\
\mathbf{Fib}_A(E_1, \Delta_p^{opl}(F_1)) \arrow[r, "\sharp_{E_1,F_1}"'] \arrow[d, "\mathbf{Fib}_A(E_{21},\Delta_p^{opl}(F_{12}))"'] \arrow[dr, phantom, "\Downarrow \sharp_{E_{21},F_{12}}"] & \mathbf{Fib}_B(\Sigma_p^{opl}(E_1), F_1) \arrow[d, "\mathbf{Fib}_B(\Sigma_p^{opl}(E_{21}),F_{12})"] \\
\mathbf{Fib}_A(E_2, \Delta_p^{opl}(F_2)) \arrow[r, "\sharp_{E_2,F_2}"'] & \mathbf{Fib}_B(\Sigma_p^{opl}(E_2), F_2)
\end{tikzcd}$$

1) Transpose functors: Let $E \in \mathbf{Fib}_A$ and $F \in \mathbf{Fib}_B$ be fibred categories. In this section, we will define the transposition functor $\sharp_{E,F} \colon \mathbf{Fib}_A(E, \Delta_p^{opl}(F)) \to \mathbf{Fib}_B(\Sigma_p^{opl}(E), F)$ and show that it is an equivalence.

a) Action of transposition on objects: Let $\phi \colon E \to \Delta_p^{opl}(F)$ be a fibred functor over A. We first define a displayed functor $\phi^\sharp \colon \Sigma_p^{opl}(E) \to F$ over B acting as follows:

- *On objects.* Fix a displayed object $\langle a, e \rangle \in \Sigma_p^{opl}(E)$ over $b \in B$, so that e lies over a and $p(a) = b$. We define $\phi_b^\sharp \langle a, e \rangle$ to be the component $\phi_a(e)\langle a, 1_a \rangle$.
- *On arrows.* Fix a displayed arrow $\langle a_{01}, e_{01} \rangle$ in $\Sigma_p^{opl}(E)$ over $b_{01} \colon b_0 \to b_1$ so that $p(a_{01}) = b_{01}$ and e_{01} lies over a_{01}. We define $\phi_{b_{01}}^\sharp \langle a_{01}, e_{01} \rangle \colon \phi_{a_0}(e_0)\langle a_0, 1_{a_0} \rangle \to \phi_{a_1}(e_1)\langle a_1, 1_{a_1} \rangle$ as follows:
  - Using the displayed functorial action of ϕ, we have a displayed arrow $\phi_{a_{01}}(e_{01}) \colon \phi_{a_0}(e_0) \to \phi_{a_1}(e_1)$ in $\Delta_p^{opl}(F)$ over a_{01}.
  - The underlying 2-cell of the above is thus an arrow $\phi_{a_{01}}(e_{01}) \colon \phi_{a_0}(e_0) \to \phi_{a_1}(e_1) \circ \Sigma_p^{opl}(\mathbf{y}_A(a_{01}))$ in $\mathbf{Fib}_B(\Sigma_p^{opl}(\mathbf{y}_A(a_0)), F)$. Instantiating this at the displayed object $\langle a_0, 1_{a_0} \rangle$ in $\Sigma_p^{opl}(\mathbf{y}_A(a_0))$ over b_0, we obtain $\phi_{a_{01}}(e_{01})\langle a_0, 1_{a_0} \rangle \colon \phi_{a_0}(e_0)\langle a_0, 1_{a_0} \rangle \to \phi_{a_1}(e_1)\langle a_0, a_{01} \rangle$.
  - Finally, we compose with the functorial action of $\phi_{a_1}(e_1)$ on the terminal map $!_{\langle a_0, a_{01} \rangle} \colon \langle a_0, a_{01} \rangle \to \langle a_1, 1_{a_1} \rangle$ in $\Sigma_p^{opl}(\mathbf{y}_A(a_1))$ over $b_{01} \colon b_0 \to b_1$.

All in all, we define $\phi_{b_{01}}^\sharp \langle a_{01}, e_{01} \rangle = \phi_{a_1}(e_1)(!_{\langle a_0, a_{01} \rangle}) \circ \phi_{a_{01}}(e_{01})\langle a_0, 1_{a_0} \rangle$.

Lemma 8: The displayed functor $\phi^\sharp \colon \Sigma_p^{opl}(E) \to F$ is fibred, *i.e.* sends cartesian arrows to cartesian arrows.

Proof: We first fix the data of a cartesian arrow in $\Sigma_p^{opl}(E)$.

$$\begin{tikzcd}
e_0 \arrow[r, "e_{01}"] \arrow[d, mapsto] \arrow[dr, phantom, "\lrcorner"] & e_1 \arrow[d, mapsto] \\
a_0 \arrow[r, "a_{01}"] \arrow[d, mapsto] \arrow[dr, phantom, "\lrcorner"] & a_1 \arrow[d, mapsto] \\
b_0 \arrow[r, "b_{01}"'] & b_1
\end{tikzcd}$$

We wish to check that the composite $\phi_{b_{01}}^\sharp \langle a_{01}, e_{01} \rangle = \phi_{a_1}(e_1)(!_{\langle a_0, a_{01} \rangle}) \circ \phi_{a_{01}}(e_{01})\langle a_0, 1_{a_0} \rangle$ is cartesian over b_{01}.

$$\begin{tikzcd}
\phi_{a_0}(e_0)\langle a_0, 1_{a_0} \rangle \arrow[r] \arrow[rr, bend left, "\phi_{b_{01}}^\sharp \langle a_{01}, e_{01} \rangle"] \arrow[d, mapsto] & \phi_{a_1}(e_1)\langle a_0, a_{01} \rangle \arrow[r] \arrow[d, mapsto] \arrow[dr, phantom, "\lrcorner"] & \phi_{a_1}(e_1)\langle a_1, 1_{a_1} \rangle \arrow[d, mapsto] \\
b_0 \arrow[r, equal] & b_0 \arrow[r, "b_{01}"'] & b_1
\end{tikzcd}$$

We first observe that the universal map $!_{\langle a_0, a_{01} \rangle} \colon \langle a_0, a_{01} \rangle \to \langle a_1, 1_{a_1} \rangle$ is cartesian in $\Sigma_p^{opl}(\mathbf{y}_A(a_1))$ over $b_{01} \colon b_0 \to b_1$ because a_{01} is cartesian over b_{01}:

$$\begin{tikzcd}
\langle a_0, a_{01} \rangle \arrow[r, "!_{\langle a_0, a_{01} \rangle}"] \arrow[d, mapsto] \arrow[dr, phantom, "\lrcorner"] & \langle a_1, 1_{a_1} \rangle \arrow[d, mapsto] \\
b_0 \arrow[r, "b_{01}"'] & b_1
\end{tikzcd}$$

Because $\phi_{a_1}(e_1) \colon \Sigma_p^{opl}(\mathbf{y}_A(a_1)) \to F$ is a fibred functor, the induced functorial arrow below is likewise cartesian in F over $b_{01} \colon b_0 \to b_1$:

$$\begin{tikzcd}
\phi_{a_1}(e_1)\langle a_0, a_{01} \rangle \arrow[r, "\phi_{a_1}(e_1)(!_{\langle a_0, a_{01} \rangle})"] \arrow[d, mapsto] \arrow[dr, phantom, "\lrcorner"] & \phi_{a_1}(e_1)\langle a_1, 1_{a_1} \rangle \arrow[d, mapsto] \\
b_0 \arrow[r, "b_{01}"'] & b_1
\end{tikzcd}$$

Because $\phi\colon E \to \Delta_p^{opl}(F)$ is itself a fibred functor over A, the following must be cartesian in $\Delta_p^{opl}(F)$ over a_{01}:

$$\begin{tikzcd} \phi_{a_0}(e_0) \arrow[r, "\phi_{a_{01}}(e_{01})"] \arrow[d, mapsto] \arrow[dr, phantom, "\lrcorner"] & \phi_{a_1}(e_1) \arrow[d, mapsto] \\ a_0 \arrow[r, "a_{01}"'] & a_1 \end{tikzcd}$$

Recall that we have split $\Delta_p^{opl}(F)$ explicitly by setting $\ell_{\Delta_p^{opl}(A)}(a_{01}, \phi_{a_1}(e_1)) = \phi_{a_1}(e_1) \circ \Sigma_p^{opl}(\mathbf{y}_A(a_{01}))$. Therefore, we have the following vertical cartesian isomorphism in $\Delta_p^{opl}(F)$ where the composite cartesian arrow is, at the level of fibred natural transformations, the identity cell:

$$\begin{tikzcd} \phi_{a_1}(e_1) \circ \Sigma_p^{opl}(\mathbf{y}_A(a_{01})) \arrow[d, dashed, "\cong"'] \arrow[dr, "\bar{\ell}_{\Delta_p^{opl}(F)}(a_{01}, \phi_{a_1}(e_1))"] & \\ \phi_{a_0}(e_0) \arrow[r, "\phi_{a_{01}}(e_{01})"] \arrow[d, mapsto] \arrow[dr, phantom, "\lrcorner"] & \phi_{a_1}(e_1) \arrow[d, mapsto] \\ a_0 \arrow[r, "a_{01}"'] & a_1 \end{tikzcd}$$

Instantiating the triangle above at $\langle a_0, 1_{a_0} \rangle$ we therefore have the following in F:

$$\begin{tikzcd} \phi_{a_1}(e_1)\langle a_0, a_{01}\rangle \arrow[d, "\cong"'] \arrow[dr, equal] & \\ \phi_{a_0}(e_0)\langle a_0, 1_{a_0}\rangle \arrow[r, "\phi_{a_{01}}(e_{01})(a_0, 1_{a_0})"'] & \phi_{a_1}(e_1)\langle a_0, a_{01}\rangle \end{tikzcd}$$

To see that $\phi^\sharp_{b_0 1}\langle a_{01}, e_{01}\rangle$ is cartesian, it is enough to see that its precomposition by the vertical cartesian isomorphism $\phi_{a_1}(e_1)\langle a_0, a_{01}\rangle \cong \phi_{a_0}(e_0)\langle a_0, 1_{a_0}\rangle$ is cartesian. But this composite is equal to the cartesian arrow $\phi_{a_1}(e_1)(!_{\langle a_0, a_{01}\rangle})$ so we are done. ■

b) Action of transposition on arrows: Let $\phi_{01}\colon \phi_0 \to \phi_1$ be a natural transformation of fibred functors $\phi_0, \phi_1\colon E \to \Delta_p^{opl}(F)$ over A. We shall define a natural transformation $\phi^\sharp_{01}\colon \phi^\sharp_0 \to \phi^\sharp_1$ in $\mathbf{Fib}_B(\Sigma_p^{opl}(E), F)$. Given a displayed object $\langle a, e\rangle \in \Sigma_p^{opl}(E)$ over $b \in B$, so that e lies over a and $p(a) = b$, we define the component $(\phi^\sharp_{01})_b^{\langle a,e\rangle}\colon (\phi^\sharp_0)_b\langle a, e\rangle \to (\phi^\sharp_1)_b\langle a, e\rangle$ as follows:

$$(\phi^\sharp_{01})_b^{\langle a,e\rangle}\colon (\phi_0)_b(e)\langle a, 1_a\rangle \to (\phi_1)_b(e)\langle a, 1_a\rangle$$
$$(\phi^\sharp_{01})_b^{\langle a,e\rangle} = (\phi_{01})_b(e)\langle a, 1_a\rangle$$

Naturality of this assignment follows from that of ϕ_{01}.

We have defined a functor $\sharp_{E,F}\colon \mathbf{Fib}_A(E, \Delta_p^{opl}(F)) \to \mathbf{Fib}_B(\Sigma_p^{opl}(E), F)$, setting $\sharp_{E,F}(\phi) :\equiv \phi^\sharp$.

2) Pseudo-naturality of transposition: The assignment $(E, F) \mapsto \sharp_{E,F}$ can be extended to a pseudo-natural transformation in $\mathbf{Fib}_A^{op} \times \mathbf{Fib}_B$. Luckily, each of the following pseudo-naturality squares commutes, in fact, on the nose:

$$\begin{tikzcd} \mathbf{Fib}_A(E_0, \Delta_p^{opl}(F_0)) \arrow[r, "\sharp_{E_0,F_0}"] \arrow[d, "\mathbf{Fib}_A(E_{10}, \Delta_p^{opl}(F_{01}))"'] & \mathbf{Fib}_B(\Sigma_p^{opl}(E_0), F_0) \arrow[d, "\mathbf{Fib}_B(\Sigma_p^{opl}(E_{10}), F_{01})"'] \\ \mathbf{Fib}_A(E_1, \Delta_p^{opl}(F_1)) \arrow[r, "\sharp_{E_1,F_1}"'] & \mathbf{Fib}_B(\Sigma_p^{opl}(E_1), F_1) \end{tikzcd}$$

This is immediate by unraveling definitions. Thus, the coherence laws for pseudo-naturality data hold as well.

3) Pseudo-invertibility of transposition: Let $E \in \mathbf{Fib}_A$ and $F \in \mathbf{Fib}_B$ be fibred categories. In this section, we will show that the transpose functor $\sharp_{E,F}\colon \mathbf{Fib}_A(E, \Delta_p^{opl}(F)) \to \mathbf{Fib}_B(\Sigma_p^{opl}(E), F)$ is an equivalence of categories. It is worth noting that the pseudo-inverse is *not* a strict inverse; this is the reason why we ultimately get a pseudo-adjunction rather than a 2-adjunction.

a) Action of reverse transposition on objects: Given a fibred functor $\phi\colon \Sigma_p^{opl}(E) \to F$ over B, we define a displayed functor $\phi^\flat\colon E \to \Delta_p^{opl}(F)$ over A acting as follows:

- *On objects.* Given $e \in E$ over $a \in A$, we define a fibred functor $\phi^\flat(e)\colon \Sigma_p^{opl}(\mathbf{y}_A(a)) \to F$ as follows:
  - *On objects.* Given $\langle a', f\rangle \in \Sigma_p^{opl}(\mathbf{y}_A(a))$, we define $\phi^\flat(e)\langle a', f\rangle = \phi\langle a', \ell_E(f, e)\rangle$.
  - *On arrows.* Given $\langle a'_{01}, f_{01}\rangle\colon \langle a'_0, f_0\rangle \to \langle a'_1, f_1\rangle$ in $\Sigma_p^{opl}(\mathbf{y}_A(a))$ over $b'_{01}\colon b'_0 \to b'_1$,[3] we define the functorial action $\phi^\flat(e)\langle a'_{01}, f_{01}\rangle\colon \phi\langle a'_0, \ell_E(f_0, e)\rangle \to \phi\langle a'_1, \ell_E(f_1, e)\rangle$ to be the functorial action of ϕ on the arrow $\langle a'_{01}, \ell_E(f_{01}, e)\rangle\colon \langle a'_0, \ell_E(f_0, e)\rangle \to \langle a'_1, \ell_E(f_1, e)\rangle$ where $\ell_E(f_{01}, e)$ is the following canonical comparison map:

$$\begin{tikzcd} \ell_E(f_0, e) \arrow[r, dashed, "\ell_E(f_{01}, e)"'] \arrow[rr, bend left, "\bar{\ell}_E(f_0, e)"] \arrow[d, mapsto] \arrow[dr, phantom, "\lrcorner"] & \ell_E(f_1, e) \arrow[r, "\bar{\ell}_E(f_1, e)"'] \arrow[d, mapsto] \arrow[dr, phantom, "\lrcorner"] & e \arrow[d, mapsto] \\ a'_0 \arrow[r, "a'_{01}"'] & a'_1 \arrow[r, "f_1"'] & a \end{tikzcd}$$

    In other words, we have defined $\phi^\flat(e)\langle a'_{01}, f_{01}\rangle = \phi(\ell_E(f_{01}, e))$.
  - *On cartesian arrows.* Let $\langle a'_{01}, f_{01}\rangle\colon \langle a'_0, f_0\rangle \to \langle a'_1, f_1\rangle$ be a cartesian arrow in $\Sigma_p^{opl}(\mathbf{y}_A(a))$, which amounts to the following data:

$$\begin{tikzcd} f_0 \arrow[r, "f_{01}"] \arrow[d, mapsto] \arrow[dr, phantom, "\lrcorner"] & f_1 \arrow[d, mapsto] \\ a'_0 \arrow[r, "a'_{01}"] \arrow[d, mapsto] \arrow[dr, phantom, "\lrcorner"] & a'_1 \arrow[d, mapsto] \\ b'_0 \arrow[r, "b'_{01}"'] & b'_1 \end{tikzcd}$$

    To see that $\phi^\flat(e)\langle a'_{01}, f_{01}\rangle = \phi(\ell_E(f_{01}, e))$ is a cartesian arrow, we recall that $\phi\colon \Sigma_p^{opl}(E) \to F$ is fibred, so it suffices to observe that the comparison map $\ell_E(f_{01}, e)\colon \ell_E(f_0, e) \to \ell_E(f_1, e)$ is cartesian by the horizontal pasting lemma for cartesian arrows.
- *On arrows.* Given $e_{01}\colon e_0 \to e_1$ over $a_{01}\colon a_0 \to a_1$, we define $\phi^\flat(e_{01})\colon \phi^\flat(e_0) \to \phi^\flat(e_1)$ in $\Delta_p^{opl}(F)$ over a_{01} as follows. Recalling the definition of displayed

[3] Here, f_{01} is the "name" of the triangle witnessing that $f_1 \circ a'_{01} = f_0$.

arrows in $\Delta_p^{opl}(F)$ we must construct a displayed natural transformation $\phi^\flat(e_{01})\colon \phi^\flat(e_0) \to \phi^\flat(e_1) \circ \Sigma_p^{opl}(\mathbf{y}_A(a_{01}))$ in $\mathbf{Fib}_B(\Sigma_p^{opl}(\mathbf{y}_A(a_0)), F)$. Given $\langle a', f\rangle \in \Sigma_p^{opl}(\mathbf{y}_A(a_0))$ so that we have $f\colon a' \to a_0$, we consider the following vertical comparison map between cartesian lifts:

$$\begin{tikzcd}
\ell_E(f, e_0) \arrow[r, "\bar{\ell}_E(f,e_0)"] \arrow[d, dashed, "\ell_E(f,e_{01})"'] & e_0 \arrow[dr, "e_{01}"] & \\
\ell_E(a_{01} \circ f, e_1) \arrow[r, "\bar{\ell}_E(a_{01}\circ f, e_1)"] \arrow[d, mapsto] \arrow[dr, phantom, "\lrcorner"] & & e_1 \arrow[d, mapsto] \\
a' \arrow[r, "f"'] & a_0 \arrow[r, "a_{01}"'] & a_1
\end{tikzcd}$$

Using the above, we define the desired functorial arrow as follows:

$$\phi^\flat(e_{01})_{\langle a', f\rangle}\colon \phi\langle a', \ell_E(f, e_0)\rangle \to \phi\langle a', \ell_E(a_{01} \circ f, e_1)\rangle$$
$$\phi^\flat(e_{01})_{\langle a', f\rangle} = \phi\langle 1_{a'}, \ell_E(f, e_{01})\rangle$$

The assignment above is natural.

Proof: To check naturality, fix $\langle a'_{01}, f_{01}\rangle\colon \langle a'_0, f_0\rangle \to \langle a'_1, f_1\rangle$ in $\Sigma_p^{opl}(\mathbf{y}_A(a))$ to check that the following square commutes in F:

$$\begin{tikzcd}
\phi\langle a'_0, \ell_E(f_0, e_0)\rangle \arrow[r, "\phi\langle 1_{a'_0}, \ell_E(f_0,e_{01})\rangle"] \arrow[d, dash, "\phi\langle a'_{01}, \ell_E(f_{01}, e_0)\rangle"'] & \phi\langle a'_0, \ell_E(a_{01} \circ f_0, e_1)\rangle \arrow[d, dash, "\phi\langle a'_{01}, \ell_E(f_{01}, e_1)\rangle"'] \\
\phi\langle a'_1, \ell_E(f_1, e_0)\rangle \arrow[r, "\phi\langle 1_{a'_1}, \ell_E(f_1,e_{01})\rangle"'] & \phi\langle a'_1, \ell_E(a_{01} \circ f_1, e_1)\rangle
\end{tikzcd}$$

It suffices to check that the following commutes in E:

$$\begin{tikzcd}
\ell_E(f_0, e_0) \arrow[r, "\ell_E(f_0,e_{01})"] \arrow[d, dash, "\ell_E(f_{01}, e_0)"'] & \ell_E(a_{01} \circ f_0, e_1) \arrow[d, dash, "\ell_E(a_{01}\cdot f_{01}, e_1)"'] \\
\ell_E(f_1, e_0) \arrow[r, "\ell_E(f_1,e_{01})"'] & \ell_E(a_{01} \circ f_1, e_1)
\end{tikzcd}$$

By the universal property of $\ell_E(a_{01} \circ f, e_1)$ as a cartesian lift, it suffices to check that the whiskering of the diagram above with $\bar{\ell}_E(a_{01} \circ f_1, e_1)\colon \ell_E(a_{01} \circ f_1, e_1) \to e_1$ commutes. By unraveling definitions and calculating, the upper and lower composites can both be simplified to $e_{01} \circ \bar{\ell}_E(f_0, e_0)\colon \ell_E(f_0, e_0) \to e_1$. ■

- *On cartesian arrows.* Now fix a *cartesian* arrow $e_{01}\colon e_0 \to e_1$ in E over $a_{01}\colon a_0 \to a_1$ in A:

$$\begin{tikzcd}
e_0 \arrow[r, "e_{01}"] \arrow[d, mapsto] \arrow[dr, phantom, "\lrcorner"] & e_1 \arrow[d, mapsto] \\
a_0 \arrow[r, "a_{01}"'] & a_1
\end{tikzcd}$$

We must check that $\phi^\flat(e_{01})\colon \phi^\flat(e_0) \to \phi^\flat(e_1)$ is cartesian in $\Delta_p^{opl}(F)$. This is the same as the underlying natural transformation $\phi^\flat(e_{01})\colon \phi^\flat(e_0) \to \phi^\flat(e_1) \circ \Sigma_p^{opl}(\mathbf{y}_A(a_{01}))$ being invertible. Fixing $f\colon a' \to a_0$ so that we have $\langle a', f\rangle \in \Sigma_p^{opl}(\mathbf{y}_A(a_0))$, we must check that the following component in F is invertible:

$$\phi^\flat(e_{01})_{\langle a', f\rangle}\colon \phi\langle a', \ell_E(f, e_0)\rangle \to \phi\langle a', \ell_E(a_{01} \circ f, e_1)\rangle$$
$$\phi^\flat(e_{01})_{\langle a', f\rangle} = \phi\langle 1_{a'}, \ell_E(f, e_{01})\rangle$$

It will be enough to observe that the universal gap map $\ell_E(f, e_{01})\colon \ell_E(f, e_0) \to \ell_E(a_{01} \circ f, e_1)$ as defined above is invertible. But this follows immediately form the pasting lemma for cartesian squares in a fibred category:

$$\begin{tikzcd}
\ell_E(f, e_0) \arrow[r, "\bar{\ell}_E(f,e_0)"] \arrow[d, mapsto] \arrow[dr, phantom, "\lrcorner"] & e_0 \arrow[r, "e_{01}"] \arrow[d, mapsto] \arrow[dr, phantom, "\lrcorner"] & e_1 \arrow[d, mapsto] \\
a' \arrow[r, "f"'] & a_0 \arrow[r, "e_{01}"'] & a_1
\end{tikzcd}$$

b) Transposition is an equivalence: We now show that each transpose functor $\sharp_{E,F}\colon \mathbf{Fib}_A(E, \Delta_p^{opl}(F)) \to \mathbf{Fib}_B(\Sigma_p^{opl}(E), F)$ is an equivalence.

Lemma 9: The transpose functor $\sharp_{E,F}$ is full.

Proof: Fix fibred functors $\phi_0, \phi_1\colon E \to \Delta_p^{opl}(F)$ over A, and a fibred natural transformation $\hat{\phi}_{01}\colon \phi_0^\sharp \to \phi_1^\sharp$. We must find $\phi_{01}\colon \phi_0 \to \phi_1$ such that $\phi_{01}^\sharp = \hat{\phi}_{01}$.

Fixing $e \in E$ over a and $f\colon a' \to a$, we must define $(\phi_{01}^e)_{\langle a', f\rangle}\colon \phi_0(e)\langle a', f\rangle \to \phi_1(e)\langle a', f\rangle$. Because each $\phi_i(e)$ is a fibred functor, the following is cartesian as $\langle a', f\rangle$ is the cartesian lift $\ell_{\Sigma_p^{opl}(\mathbf{y}_A(a))}(f, \langle a, 1_a\rangle)$.

$$\begin{tikzcd}
\phi_i(e)\langle a', f\rangle \arrow[r, "\phi_i(e)(1_{\langle a', f\rangle})"] \arrow[d, mapsto] & \phi_i(e)\langle a, 1_a\rangle \arrow[d, mapsto] \\
a' \arrow[r, "f"'] & a
\end{tikzcd}$$

We already have $\hat{\phi}_{01}^{\langle a,e\rangle}\colon \phi_0(e)\langle a, 1_a\rangle \to \phi_1(e)\langle a, 1_a\rangle$ by assumption. We can use this together with the universal property of the cartesian arrow depicted above to define $(\phi_{01}^e)_{\langle a', f\rangle}\colon \phi_0(e)\langle a', f\rangle \to \phi_1(e)\langle a', f\rangle$ as follows:

$$\begin{tikzcd}
\phi_0(e)\langle a', f\rangle \arrow[r, "\phi_0(e)(1_{\langle a', f\rangle})"] \arrow[d, dashed, "(\phi_{01}^e)_{\langle a', f\rangle}"'] & \phi_0(e)\langle a, 1_a\rangle \arrow[d, "\hat{\phi}_{01}^{\langle a,e\rangle}"] \\
\phi_1(e)\langle a', f\rangle \arrow[r, "\phi_1(e)(1_{\langle a', f\rangle})"'] & \phi_1(e)\langle a, 1_a\rangle
\end{tikzcd}$$

We must check that $\phi_{01}^\sharp = \hat{\phi}_{01}$. Fixing $\langle a, e\rangle \in \Sigma_p^{opl}(E)$ over b so that e lies over a and $p(a) = b$, we must check that $(\phi_{01}^e)_{\langle a, 1_a\rangle} = \hat{\phi}_{01}^{\langle a,e\rangle}$. By the universal property of the cartesian lift, it suffices to check that the following square commutes:

$$\begin{tikzcd}
\phi_0(e)\langle a, 1_a\rangle \arrow[r, "\phi_0(e)(1_{\langle a, 1_a\rangle})"] \arrow[d, "\hat{\phi}_{01}^{\langle a,e\rangle}"'] & \phi_0(e)\langle a, 1_a\rangle \arrow[d, "\hat{\phi}_{01}^{\langle a,e\rangle}"] \\
\phi_1(e)\langle a, 1_a\rangle \arrow[r, "\phi_1(e)(1_{\langle a, 1_a\rangle})"'] & \phi_1(e)\langle a, 1_a\rangle
\end{tikzcd}$$

But this is a naturality square for $\hat{\phi}_{01}\colon \phi_0^\sharp \to \phi_1^\sharp$. ■

Lemma 10: The transpose functor $\sharp_{E,F}$ is faithful.

Proof: We fix a pair of displayed natural transformations $\phi_{01}, \psi'_{01} \colon \phi_0 \to \phi_1$ between fibred functors $\phi_0, \phi_1 \colon E \to \Delta_p^{opl}(F)$ over A such that $\phi_{01}^\sharp = \psi_{01}^\sharp$. We need to check that $\phi_0 = \phi_1$, *i.e.* that for any $e \in E$ over $a \in A$ we have $\phi_{01}(e) = \psi_{01}(e)$ in $\Delta_p^{opl}(F)$. Fixing $\langle a', f\rangle \in \Sigma_p^{opl}(\mathbf{y}_A(a))$ over $p(a')$, we must check that $\phi_{01}(e)\langle a', f\rangle = \psi_{01}(e)\langle a', f\rangle$.

But $\langle a', f\rangle$ is the cartesian lift $\ell_{\Sigma_p^{opl}(\mathbf{y}_A(a))}(f, \langle a, 1_a\rangle)$. Because $\phi_0(e)$ and $\phi_1(e)$ are fibred functors, the following are cartesian:

$$\begin{tikzcd}
\phi_i(e)\langle a', f\rangle \arrow[r, "\phi_i(e)(1_{\langle a', f\rangle})"] \arrow[d, mapsto] & \phi_i(e)\langle a, 1_a\rangle \arrow[d, mapsto] \\
a' \arrow[r, "f"'] & a
\end{tikzcd}$$

and thus in the following naturality square, $\phi_{01}(e)\langle a', f\rangle$ is the *unique* vertical map that can be configured like so:

$$\begin{tikzcd}
\phi_0(e)\langle a', f\rangle \arrow[r, "\phi_0(e)(1_{\langle a', f\rangle})"] \arrow[d, "\phi_{01}(e)\langle a', f\rangle"'] & \phi_0(e)\langle a, 1_a\rangle \arrow[d, "\phi_{01}(e)\langle a, 1_a\rangle"] \\
\phi_1(e)\langle a', f\rangle \arrow[r, "\phi_1(e)(1_{\langle a', f\rangle})"'] & \phi_1(e)\langle a, 1_a\rangle
\end{tikzcd}$$

Thus, it suffices to check that the following square commutes:

$$\begin{tikzcd}
\phi_0(e)\langle a', f\rangle \arrow[r, "\phi_0(e)(1_{\langle a', f\rangle})"] \arrow[d, "\psi_{01}(e)\langle a', f\rangle"'] & \phi_0(e)\langle a, 1_a\rangle \arrow[d, "\phi_{01}(e)\langle a, 1_a\rangle"] \\
\phi_1(e)\langle a', f\rangle \arrow[r, "\phi_1(e)(1_{\langle a', f\rangle})"'] & \phi_1(e)\langle a, 1_a\rangle
\end{tikzcd}$$

Becuase $\phi_{01}^\sharp = \psi_{01}^\sharp$, we have $\phi_{01}(e)\langle a, 1_a\rangle = \psi_{01}(e)\langle a, 1_a\rangle$; thus is suffices to check that the following square commmutes:

$$\begin{tikzcd}
\phi_0(e)\langle a', f\rangle \arrow[r, "\phi_0(e)(1_{\langle a', f\rangle})"] \arrow[d, "\psi_{01}(e)\langle a', f\rangle"'] & \phi_0(e)\langle a, 1_a\rangle \arrow[d, "\psi_{01}(e)\langle a, 1_a\rangle"] \\
\phi_1(e)\langle a', f\rangle \arrow[r, "\phi_1(e)(1_{\langle a', f\rangle})"'] & \phi_1(e)\langle a, 1_a\rangle
\end{tikzcd}$$

This is a naturality square for $\psi_{01}(e)$, so we are done. ∎

Lemma 11: The reverse transpose operation exhibits the transpose functor $\sharp_{E,F}$ as *(split) essentially surjective*.

Proof: Fixing a fibred functor $\phi \colon \Sigma_p^{opl}(E) \to F$ over B, we will check that $\phi^{\flat\sharp}$ is naturally isomorphic to ϕ. By definition, we have the following computation of the component of $\phi^{\flat\sharp}$ at $\langle a, e\rangle \in \Sigma_p^{opl}(E)$ over $b \in B$:

$$\phi_b^{\flat\sharp}\langle a, e\rangle = \phi_a^\flat(e)\langle a, 1_a\rangle = \phi_b\langle a, \ell_E(1_a, e)\rangle$$

The cartesian arrow $\bar{\ell}_E(1_a, e) \colon \ell_E(1_a, e) \to e$ is an isomorphism because 1_a is an identity arrow, so we have an isomorphism $\phi_{1_b}\langle 1_a, \bar{\ell}_E(1_a, e)\rangle \colon \phi_b^{\flat\sharp}\langle a, e\rangle \to \phi_b\langle a, e\rangle$ by functoriality. For naturality, we fix $e_{01} \colon e_0 \to e_1$ over $a_{01} \colon a_0 \to a_1$ over $b_{01} \colon b_0 \to b_1$ to check that the following commutes in F:

$$\begin{tikzcd}
\phi_{b_0}^{\flat\sharp}\langle a_0, e_0\rangle \arrow[r, "\phi_{b_01}^{\flat\sharp}\langle a_{01}, e_{01}\rangle"] \arrow[d, dashed, "\phi_{1_{b_0}}\langle 1_{a_0}, \bar{\ell}_E(1_{a_0}, e_0)\rangle"'] & \phi_{b_1}^{\flat\sharp}\langle a_1, e_1\rangle \arrow[d, dashed, "\phi_{1_{b_1}}\langle 1_{a_1}, \bar{\ell}_E(1_{a_1}, e_1)\rangle"] \\
\phi_{b_0}\langle a_0, e_0\rangle \arrow[r, "\phi_{b_01}\langle a_{01}, e_{01}\rangle"'] & \phi_{b_1}\langle a_1, e_1\rangle
\end{tikzcd}$$

We begin with the following commuting square in E:

$$\begin{tikzcd}
\ell_E(1_{a_0}, e_0) \arrow[r, "\ell_E(1_{a_0}, e_{01})"] \arrow[d, "\bar{\ell}_E(1_{a_0}, e_0)"'] & \ell_E(a_{01}, e_1) \arrow[d, "\bar{\ell}_E(a_{01}, e_1)"] \\
e_0 \arrow[r, "e_{01}"'] & e_1
\end{tikzcd}$$

Under $\phi \colon \Sigma_p^{opl}(E) \to F$, we obtain the following:

$$\begin{tikzcd}
\phi_{b_0}\langle a_0, \ell_E(1_{a_0}, e_0)\rangle \arrow[r, "\phi_{1_{b_0}}\langle 1_{a_0}, \ell_E(1_{a_0}, e_{01})\rangle"] \arrow[d, dashed, "\phi_{1_{b_0}}\langle 1_{a_0}, \bar{\ell}_E(1_{a_0}, e_0)\rangle"'] & \phi_{b_0}\langle a_0, \ell_E(a_{01}, e_1)\rangle \arrow[d, dashed, "\phi_{b_{01}}\langle a_{01}, \bar{\ell}_E(a_{01}, e_1)\rangle"] \\
\phi_{b_0}\langle a_0, e_0\rangle \arrow[r, "\phi_{b_{01}}\langle a_{01}, e_{01}\rangle"'] & \phi_{b_1}\langle a_1, e_1\rangle
\end{tikzcd}$$

By definition of the comparison map $\ell_E(!_{\langle a_0, a_{01}\rangle}, e_0)$, we have the following triangle that factors the eastern map in the square above:

$$\begin{tikzcd}
\phi_{b_0}\langle a_0, \ell_E(a_{01}, e_1)\rangle \arrow[rr, "\phi_{b_{01}}\langle a_{01}, \ell_E(!_{\langle a_0, a_{01}\rangle}, e_0)\rangle"] \arrow[dr, "\phi_{b_{01}}\langle a_{01}, \bar{\ell}_E(a_{01}, e_1)\rangle"'] & & \phi_{b_1}\langle a_1, \ell_E(1_{a_1}, e_1)\rangle \arrow[dl, "\phi_{1_{b_1}}\langle 1_{a_1}, \bar{\ell}_E(1_{a_1}, e_1)\rangle"] \\
& \phi_{b_1}\langle a_1, e_1\rangle &
\end{tikzcd}$$

Returning to the original naturality square that we wished to check, we unfold definitions to see that $\phi_{b_{01}}^{\flat\sharp}\langle a_{01}, e_{01}\rangle$ is the composite $\phi_{b_{01}}\langle a_{01}, \ell_E(!_{\langle a_0, a_{01}\rangle}, e_0)\rangle \circ \phi_{1_{b_0}}\langle 1_{a_0}, \ell_E(1_{a_0}, e_{01})\rangle$. Therefore, we have:

$$\begin{aligned}
&\phi_{1_{b_1}}\langle 1_{a_1}, \bar{\ell}_E(1_{a_1}, e_1)\rangle \circ \phi_{b_{01}}^{\flat\sharp}\langle a_{01}, e_{01}\rangle \\
&\quad = \phi_{1_{b_1}}\langle 1_{a_1}, \bar{\ell}_E(1_{a_1}, e_1)\rangle \circ \phi_{b_{01}}\langle a_{01}, \ell_E(!_{\langle a_0, a_{01}\rangle}, e_0)\rangle \\
&\qquad \circ \phi_{1_{b_0}}\langle 1_{a_0}, \ell_E(1_{a_0}, e_{01})\rangle \\
&\quad = \phi_{b_{01}}\langle a_{01}, \bar{\ell}_E(a_{01}, e_1)\rangle \circ \phi_{1_{b_0}}\langle 1_{a_0}, \ell_E(1_{a_0}, e_{01})\rangle \\
&\quad = \phi_{b_{01}}\langle a_{01}, e_{01}\rangle \circ \phi_{1_{b_0}}\langle 1_{a_0}, \bar{\ell}_E(1_{a_0}, e_0)\rangle
\end{aligned}$$

We are done. ∎

Corollary 12: The transpose functor $\sharp_{E,F}$ is a (strong) equivalence of categories.

Corollary 13: For a fibration $p \colon A \twoheadrightarrow B$, the oplax sum pseudofunctor $\Sigma_p^{opl} \colon \mathbf{Fib}_A \to \mathbf{Fib}_B$ has a right pseudo-adjoint given by $\Delta_p^{opl} \colon \mathbf{Fib}_B \to \mathbf{Fib}_A$.

C. Relative Hofmann–Streicher lifting

The (relative, generalised) Hofmann–Streicher of a fibred category $E \in \mathbf{Fib}_B$ along a fibration $p \colon A \twoheadrightarrow B$ is

finally obtained by applying the *lax* base change functor $\Delta_p^{lax}\colon \mathbf{Fib}_B \to \mathbf{Fib}_A$ to E as described in § II-F:

$$\Delta_p^{lax}(E) = \Delta_p^{opl}(E^{op})^{op}$$

As we have explained, the generalisation from ordinary categories A to fibrations $p\colon A \twoheadrightarrow B$ allows us to iterate the Hofmann–Streicher lifting construction along a sequence of fibrations $A_0 \twoheadrightarrow \cdots A_n \twoheadrightarrow B$; post-composing finally with $B \twoheadrightarrow \mathbf{1}$ we see that our construction agrees with the usual "absolute" Hofmann–Streicher lifting of a category.

IV. Conclusions and Future Work

A. Relationship to weak bisimulation in concurrency semantics

There is one additional piece of related work that deserves disussion, namely the *saturation monad* of Fiore, Cattani, and Winskel [25]. In the presheaf semantics of concurrency, a closed process is interpreted as a presheaf on a "path category" $\mathbb{P}$—such as the category of pomsets. Joyal, Nielsen, and Winskel [26] showed that *strong bisimulation* of processes has a very simple description in terms of the presheaf semantics:

1) An *open map* $f\colon A \to B$ of presheaves is a natural transformation that has the right lifting property with respect to every $\mathbf{y}_{\mathbb{P}}(x) \to \mathbf{y}_{\mathbb{P}}(y)$.
2) A *strong bisimulation* between processes A and B is a span $A \leftarrow C \to B$ of open maps.

Open map bisimulation corresponds exactly to *real* strong bisimulation within the image of various full embeddings of concrete concurrency models into $\hat{\mathbb{P}}$. The result of Fiore, Cattani, and Winskel [25] is to reduce *weak* bisimulation to (strong) open map bisimulation using a saturation monad on presheaves, by analogy with existing saturation monads on concrete concurrency models like labelled transition systems.

1) The 1-dimensional saturation adjunction: For a given hiding functor $h\colon \mathbb{P} \to \mathbb{Q}$, Fiore *et al.*'s saturation monad on $\hat{\mathbb{P}}$ is obtained from the following 1-categorical adjunction, where we have again written $\mathbf{cat}_s$ for the 1-category of small 1-categories:

$$\begin{tikzcd} \hat{\mathbb{P}} \arrow[r, hook, shift left] \arrow[r, phantom, "\perp"] & \mathbf{cat}_s/\mathbb{P} \arrow[l, shift left] \arrow[r, shift left, "\Sigma_h"] \arrow[r, phantom, "\perp"] & \mathbf{cat}_s/\mathbb{Q} \arrow[l, shift left, "\Delta_h"] \end{tikzcd}$$

The left adjoint $\hat{\mathbb{P}} \hookrightarrow \mathbf{cat}_s/\mathbb{P}$ depicted above takes a presheaf to the display of its category of elements over $\mathbb{P}$. The composite right adjoint sends a displayed category X over $\mathbb{Q}$ to the presheaf $\mathbf{cat}_s/\mathbb{Q}(\Sigma_h(\mathbf{y}_{\mathbb{P}}(-)), X)$.

2) 2-dimensional saturation: In order to compare Fiore *et al.*'s saturation construction on an even playing field with our relative Hofmann–Streicher lifting 2-functor, we must upgrade the former to a pseudo-adjunction. First we recall the 2-adjunction $U_{\mathbb{P}} \dashv N_{\mathbb{P}}\colon \mathbf{Cat}/\mathbb{P} \to \mathbf{Fib}_{\mathbb{P}}$ that gives *co-free* fibred categories, having recently been studied by Emmenegger, Mesiti, Rosolini, and Streicher [27]. The right adjoint $N_{\mathbb{P}}$ sends a displayed category X over $\mathbb{P}$ to the fibration $\Sigma_{\partial_0}(\Delta_{\partial_1}(X))$:

$$\begin{tikzcd} \pi_X/B \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & B^{\to} \arrow[r, "\partial_0"'] \arrow[d, "\partial_1"] & B \\ X \arrow[r, "\pi_X"'] & B \arrow[ur, bend left, "N_{\mathbb{P}}(X)"] & \end{tikzcd}$$

Then the spirit of Fiore *et al.*'s saturation adjunction is most faithfully captured by the following pseudo-adjunction:

$$\begin{tikzcd} \mathbf{Fib}_{\mathbb{P}} \arrow[r, hook, shift left, "U_{\mathbb{P}}"] \arrow[r, phantom, "\perp"] & \mathbf{Cat}/\mathbb{P} \arrow[l, shift left, "N_{\mathbb{P}}"] \arrow[r, shift left, "\Sigma_h"] \arrow[r, phantom, "\perp"] & \mathbf{Cat}/\mathbb{Q} \arrow[l, shift left, "\Delta_h"] \end{tikzcd}$$

A displayed category $X \in \mathbf{Cat}/\mathbb{Q}$ is sent by the right pseudo-adjoint above to the (split) fibred category corresponding under straightening to the following diagram of categories:

$$N_{\mathbb{P}}(\Delta_h(X))_\bullet \simeq \mathbf{Cat}/\mathbb{Q}(\Sigma_h(\mathbf{y}_{\mathbb{P}}(-), X))$$

3) Comparing saturation with relative Hofmann–Streicher lifting: We are now able to compare the 2-dimensional version of the saturation adjunction with our relative Hofmann–Streicher lifting adjunction. We will see that the two constructions agree only on *co-free* fibrations below.

Observation 14: When $h\colon \mathbb{P} \twoheadrightarrow \mathbb{Q}$ is a fibration, the upgraded right pseudo-adjoint $N_{\mathbb{P}} \circ \Delta_h\colon \mathbf{Cat}/\mathbb{Q} \to \mathbf{Fib}_{\mathbb{P}}$ of Fiore *et al.* factors through our own $\Delta_h^{opl}\colon \mathbf{Fib}_{\mathbb{Q}} \to \mathbf{Fib}_{\mathbb{P}}$ up to equivalence as follows:

$$\begin{tikzcd} \mathbf{Cat}/\mathbb{Q} \arrow[r, "N_{\mathbb{Q}}"] \arrow[d, "\Delta_h"'] & \mathbf{Fib}_{\mathbb{Q}} \arrow[d, "\Delta_h^{opl}"] \\ \mathbf{Cat}/\mathbb{P} \arrow[r, "N_{\mathbb{P}}"'] & \mathbf{Fib}_{\mathbb{P}} \end{tikzcd}$$

Sketch: Letting $X \in \mathbf{Cat}/\mathbb{Q}$ be a displayed category, we compute as follows:

$$\begin{aligned} &\Delta_h^{opl}(N_{\mathbb{Q}}(X))_\bullet \\ &\quad \simeq \mathbf{Fib}_{\mathbb{P}}(\mathbf{y}_{\mathbb{P}}(-), \Delta_h^{opl}(N_{\mathbb{Q}}(X))) \\ &\quad \simeq \mathbf{Fib}_{\mathbb{Q}}(\Sigma_h^{opl}(\mathbf{y}_{\mathbb{P}}(-)), N_{\mathbb{Q}}(X)) \\ &\quad \simeq \mathbf{Cat}/\mathbb{Q}(U_{\mathbb{Q}}(\Sigma_h^{opl}(\mathbf{y}_{\mathbb{P}}(-))), X) \\ &\quad \simeq \mathbf{Cat}/\mathbb{Q}(\Sigma_h(\mathbf{y}_{\mathbb{P}}(-)), X) \\ &\quad \simeq N_{\mathbb{P}}(\Delta_h(X))_\bullet \end{aligned}$$

We are done. ■

Observation 15: The co-free fibration on a displayed category $X \in \mathbf{Cat}/\mathbf{1}$ over the point is precisely X, *i.e.* we have $N_{\mathbf{1}}(X) \simeq X$.

From these two observations, we immediately obtain the following.

Corollary 16: Saturation agrees with (relative) Hofmann–Streicher lifting over the point. More precisely, for any categories C and E, we have $N_{\mathbb{P}}(\Delta_{!_C}(E)) \simeq \Delta_{!_C}^{opl}(E)$.

Therefore, our construction and that of Fiore *et al.* are two *different* generalisations of Hofmann–Streicher lifting; ours is the correct one for internalising and iterating Hofmann–Streicher lifting, whereas theirs is the correct one for reducing weak bisimulation to (strong) open map bisimulation in presheaf models of concurrency. It would be very interesting indeed to find out whether our own relative Hofmann–Streicher lifting does anything useful in the world of concurrency semantics.

B. A new 2-dimensional bifibration of fibrations?

The relationship between fibred categories and their base categories is well-expressed by the 2-fibration $\mathbf{Fib} \to \mathbf{Cat}$ in which cartesian lifts of one-cells are given by pullbacks of fibrations. Hermida [28] explains that for a functor $f\colon A \to B$, each of the pullback 2-functors $f^*\colon \mathbf{Fib}_B \to \mathbf{Fib}_A$ has a left pseudo-adjoint $f_!\colon \mathbf{Fib}_A \to \mathbf{Fib}_B$; the full computation of the left pseudo-adjoint is not easy, but its fibre categories are described by the following pseudo-colimits:

$$(f_!E)_b = \mathrm{pscolim}\big((b/f)^{op} \xrightarrow{\partial_1^{op}} A^{op} \xrightarrow{E_\bullet} \mathbf{Cat}\big)$$

In summary, these left adjoints endow the 2-fibration $\mathbf{Fib} \rightarrowtriangle \mathbf{Cat}$ with *direct images* in the sense of Hermida [28], making $\mathbf{Fib} \rightarrowtriangle \mathbf{Cat}$ into some kind of 2-dimensional bifibration—although a consensus on what exactly a bifibration of 2-categories ought to be remains forthcoming.[4]

In this paper, we have shown that for each fibration $p\colon A \rightarrowtriangle B$ we have a pseudo-adjunction $\Sigma_p^{opl} \dashv \Delta_p^{opl}\colon \mathbf{Fib}_A \to \mathbf{Fib}_B$, and it is natural to ask whether these play a part in a *different* 2-dimensional bifibration (whatever that needs to mean). If this is going to be the case, we recognise at the outset that the base 2-category should have categories for 0-cells but *fibrations* $A \rightarrowtriangle B$ for 1-cells; there is not an obvious choice for a definition of 2-cells between fibrations $p, q\colon A \rightarrowtriangle B$ with the same total category.

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[4]Even the definition of "2-fibration" is contested in the literature: see Buckley [29, Remark 2.1.9] for discussion. Buckley's definition is strictly stronger than that of Hermida [30], and this strength is necessary in order to obtain the straightening/unstraightening theory of 2-fibrations as Buckley points out, apparently refuting a claim of Baković [31]. Therefore, any work that explicitly depends on the elegant results of Hermida [28], [30] on 2-fibrations would need to check explicitly whether these results are in fact compatible with Buckley's corrected definition of 2-fibrations.

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